The Labor of Alignment: How Peer Specialists Make AI Fit Peer-Support Practice

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Abstract

AI alignment efforts often specify values before use, yet making AI outputs appropriate in practice can still require substantial human work. We examine this work in peer support, where peer specialists evaluated and adapted outputs from an LLM-based assistant already configured with established peer-support values. Through a three-phase study with 18 peer specialists, we find that alignment work persisted even when AI-generated support appeared broadly consistent with these values, because peer specialists still had to determine what appropriate support required for particular service users and situations. We synthesize this work into a process model showing how peer specialists diagnosed misalignment, repaired AI-generated support, and judged its sufficiency for intended use, drawing on complementary forms of knowledge while navigating challenges throughout. We conceptualize this work as *last-mile alignment labor*, show how AI can redistribute work toward less-visible forms of evaluation and adaptation, and offer design and organizational implications for supporting it.

CCS Concepts

• **Human-centered computing → Empirical studies in HCI**.

1 Introduction

As AI systems increasingly participate in decisions, interactions, and practices that shape people's lives, they are expected not only to be capable, but to act in ways that reflect people's values and expectations. This challenge is often framed as "value alignment": shaping AI behavior to be consistent with human values and normative expectations [21]. Much work on AI value alignment has sought to shape such behavior before use through model- and system-level interventions such as preference learning [13, 46, 83], instruction tuning [14, 54, 78], and system-level instructions [4, 21, 44, 77]. For example, Constitutional AI guides language-model behavior using a set of high-level principles, or a "constitution," intended to specify desirable behavior [7]. Yet broadly specified values and principles may not fully determine appropriate AI behavior across different domains and situations of use. For example, a person who generally prioritizes freedom over safety may instead prioritize safety during a global pandemic [36].

Peer support is a domain in which this limitation becomes particularly visible. Peer specialists are people with lived experience of mental health challenges and recovery who draw on that experience to support service users navigating similar experiences and related socioeconomic challenges. On the one hand, peer support is guided by well-established values and principles, including mutuality, self-determination, hope, and person-centered relationships [45, 61, 71, 73]. As AI becomes increasingly integrated into peer-support work, these values must also guide how AI-generated support is interpreted and used in practice [52, 72]. Yet these shared values do not prescribe a single form of appropriate support. Importantly, peer specialists use AI to generate support for service users who do not interact with the system directly, requiring them to interpret what values such as autonomy, non-directiveness, and empathy require for a particular person, situation, and relationship.

Recent HCI research has shown that users actively interpret and reshape AI behaviors by prompting, editing, rejecting, and correcting outputs, drawing on their goals, values, identities, and contextual knowledge [10, 18, 26, 34, 37, 43]. Other work shows that AI aligned around broadly specified values may still fail to reflect

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the distinctive values and requirements of particular domains of practice. Comedians, clinicians, and teachers, for example, adapt general-purpose AI outputs when they conflict with the norms, judgments, or requirements of their work [25, 38, 81]. Arzberger et al. [3] further argue that predefined alignment cannot exhaust what alignment requires in situated use, showing how users continue to surface, negotiate, and respond to misalignment in context and highlighting the human labor this ongoing process can entail.

Our study examines what happens after a further step has been taken: relevant practice values have already been explicitly incorporated into the AI system. What alignment work, if any, remains under these conditions?

Understanding this remaining work connects to a longstanding HCI literature showing that making technologies work in practice often depends on substantial and frequently invisible human labor, including accommodating contingencies, repairing breakdowns, and adapting systems to local circumstances [20, 30, 57, 63, 64]. Research on AI similarly shows that ostensibly automated systems can shift verification, interpretation, maintenance, and repair work to the people who use them [16, 47, 74]. Yet we know less about the forms and processes of alignment labor that remain when practice values are already represented in an AI system and its outputs must be made appropriate for particular others and intended uses.

To address these gaps, we examine how peer specialists make AI-generated support fit particular service users and peer-support practice when the system is already configured with a set of well-established peer-support values.

We ask the following research questions:

- **RQ1.** How do peer specialists draw on knowledge to judge whether AI-generated support fits a particular situation?
- **RQ2.** How do peer specialists adapt AI-generated support when it does not fit peer-support practice?
- **RQ3.** What challenges and forms of labor arise as peer specialists evaluate and adapt AI-generated support?

To investigate these questions, we conducted a three-phase study with 18 peer specialists, consisting of baseline interviews and orientation sessions, a two-week diary study using PeerCompass, and paired follow-up interviews. PeerCompass is an LLM-based assistant designed to support peer specialists with everyday tasks such as generating personalized plans for service users, identifying next steps, and locating relevant resources. We configured the system with a prompt incorporating a set of well-established peer-support values and principles [19, 40, 65, 66]. Using de-identified situations drawn from their peer-support practice, participants documented cases in which AI-generated support did not fit the values of their practice or the needs and circumstances of the people they support, and reflected on how they addressed these mismatches. Follow-up interviews examined how participants evaluated and adapted this support, the challenges they encountered, and the forms of labor involved.

Our findings show that substantial alignment work persisted with peer specialists even when peer-support values were explicitly incorporated into the system prompt. Participants drew on complementary knowledge of peer-support practice, lived experience, and particular service users and relationships to diagnose misalignment, repair AI-generated support, and judge its sufficiency for intended use, while navigating challenges throughout this process. Together, these activities unfolded as an iterative process, which we synthesize in our process model. Building on these findings, we identify design and organizational opportunities for supporting and recognizing this alignment work.

In this study, our contributions are threefold:

- We show that even when an AI system is explicitly configured with a well-established set of practice values, making AI-generated support appropriate *for others* can still involve substantial human labor to determine what those values require for particular people and situations. Drawing on Gray and Suri's metaphor of *the last mile of automation* [23], we conceptualize this remaining work as *last-mile alignment labor*.
- We provide an empirical account of how this *last-mile alignment labor* unfolds in AI-supported peer support. We show how peer specialists draw on complementary forms of knowledge to diagnose misalignment, repair outputs, and judge sufficiency for intended use, and identify the challenges they encounter throughout this process. We synthesize this work into an iterative process model.
- We offer design and organizational implications grounded in these findings for supporting alignment work in peer-support practice.

2 Related Work

2.1 From Value Alignment During Development to Alignment in Use

Much of AI alignment research has focused on shaping model behavior around human intentions and values before use through approaches such as instruction tuning [14, 54, 78], reinforcement learning from human feedback [6, 13, 46], and constitutional supervision [7, 29, 60, 70]. These approaches use preferences, instructions, or explicit principles during model development to guide subsequent behavior across users and contexts. However, broadly specified values in advance may not fully determine what alignment requires across particular domains and situations of use [2, 3, 79].

A longstanding line of HCI research has examined how people shape AI systems when they encounter problematic behaviors in use. Users develop folk theories and adjust their interactions to influence outputs [17]; when systems conflict with their identities or values, they may push back against or "tame" them [32, 62]. Users also collectively identify problematic behavior and gather evidence through practices similar to auditing and remediation [15].

With LLMs, users iteratively prompt, accept, reject, edit, and regenerate outputs to bring them closer to their goals [10, 34, 43, 80]. These interventions extend beyond task performance. Users revise AI-generated self-representations to reflect their identities and values [26], adapt content that fails to reflect their voice and context [41], and reshape AI behavior to better reflect their values and preferences [18, 35, 37]. Fan et al. [18] conceptualize this as *user-driven value alignment*, where users actively identify, challenge, and attempt to correct AI outputs they perceive as harmful, aiming to guide the AI to better align with their values.

Other work further shows that AI aligned around broadly specified values may still fail to reflect the distinctive values and requirements of particular domains of practice. Clinicians verify and revise LLM outputs that do not satisfy clinical requirements [25, 38]; teachers modify or reject content that conflicts with pedagogical judgments or relational values [81]; and comedians adapt general-purpose AI when its behavior conflicts with the norms and values of their creative practice [41]. Arzberger et al. [3] further argue that predefined alignment cannot exhaust what alignment requires in situated use, showing how users continue to surface, negotiate, and respond to misalignment in context and highlighting the human labor this ongoing process can entail.

However, we know less about what alignment work, if any, remains when relevant practice values have already been incorporated into an AI system and AI users must make its outputs appropriate not for themselves but for particular others and situations. To address this gap, we investigate how peer specialists make AI-generated support fit particular service users and peer-support practice when the system is already configured with a set of well-established peer-support values.

2.2 The Human Labor of Making Technology Work in Practice

A longstanding body of HCI research has shown that making technologies work in practice often requires substantial human effort. Because technologies cannot anticipate all circumstances of use, people must bridge gaps between system designs and particular situations [22, 24, 57, 63, 64]. Scholars have described this effort as *articulation work*: accommodating contingencies and coordinating people, technologies, and tasks so sociotechnical arrangements continue to function [57, 63, 64]. When technologies fail to fit local circumstances, people further devise workarounds [30, 42, 48, 76]. Much of this work remains invisible when systems appear to function smoothly [27, 63, 67], and often draws on contextual, experiential, and tacit knowledge that formal system representations cannot fully capture [1, 53].

These dynamics are particularly consequential in care settings, where making technologies workable is intertwined with relational and contextual demands. Mental health helpline volunteers, for example, perform invisible work to maintain the institutional, interpersonal, and individual boundaries through which technology-mediated support becomes possible [51]. In long-term care, frontline care workers similarly collect and organize personalized sociopsychological, emotional, and ecological information that standardized systems struggle to capture [69]. Together, this work shows how making technologies workable in care depends on ongoing human efforts to accommodate particular people, relationships, and circumstances.

AI similarly does not eliminate this labor. In healthcare, workers and patients repair algorithmic representations that fail to correspond with the realities of care [58]; nurses contextualize risk scores, time alerts around clinical workflows, and adapt communication with physicians [16]; and clinicians translate misleading AI recommendations into appropriate actions and monitor unexpected model behavior [47, 74]. Beyond healthcare, Fox et al. [20] characterize the hidden human labor of AI integration as *patchwork*, through which workers calibrate, troubleshoot, and repair AI systems to bridge gaps between what AI purports to do and what it accomplishes in practice. Related work documents invisible infrastructuring around digital technologies in elder care [59], as well as often-overlooked labor underlying AI development and alignment, including system production, labeling, evaluation, and oversight [23, 28].

Together, prior research richly documents the articulation, repair, maintenance, and often invisible labor required to make technologies and AI workable in practice. Yet we know less about the *alignment labor* that remains under a different condition: relevant practice values are already incorporated into an AI system, but its outputs must still be evaluated and adapted for particular others and intended uses. We examine this remaining labor through peer specialists' evaluation and adaptation of AI-generated support for particular service users and situations.

3 Method

We conducted a three-phase study with 18 certified peer specialists to examine how they evaluated and responded to AI-generated support from a system whose prompt incorporated peer-support values, and the challenges and labor that arose. First, a baseline interview and orientation introduced the LLM-based system and study task; next, a two-week diary study captured participants' independent identification of and responses to misalignment across repeated use; finally, paired follow-up interviews allowed participants to revisit and compare cases and reflect on their judgments and experiences (see Figure 1).

3.1 PeerCompass and Peer-Support Value Configuration

3.1.1 PeerCompass. We used PeerCompass, an LLM-powered assistant designed to support peer specialists working with service users at peer-run behavioral health organizations. Its primary LLM-powered feature, *Chat + Explore*, allows peer specialists to describe a service user's situation and receive structured support for tasks including developing personalized plans, suggesting communication strategies, translating broad goals into concrete next steps, and identifying organizational and community resources. PeerCompass uses a retrieval-augmented generation pipeline over thousands of resources vetted by peer specialists. *Chat + Explore* consists of a conversational window and persistent sidebar. Peer specialists can iteratively refine AI-generated support through follow-up messages that add context or redirect the response. The sidebar organizes generated information into Active Goals and Resources, including next steps, relevant organizations, and contact or referral information.

3.1.2 Specifying Peer-Support Values in the System Prompt. Before the study, we configured PeerCompass with a system prompt incorporating a set of well-established peer-support values and principles [19, 40, 65, 66], with guidance from co-authors with expertise spanning peer support, social work, occupational therapy, behavioral health, and lived-experience advocacy. The prompt emphasized empathy, mutuality, hope, service-user autonomy, non-clinical and non-directive support, recovery-oriented practice, and appropriate role boundaries. These instructions provided an explicit, literature-grounded specification of broad peer-support values, rather than an

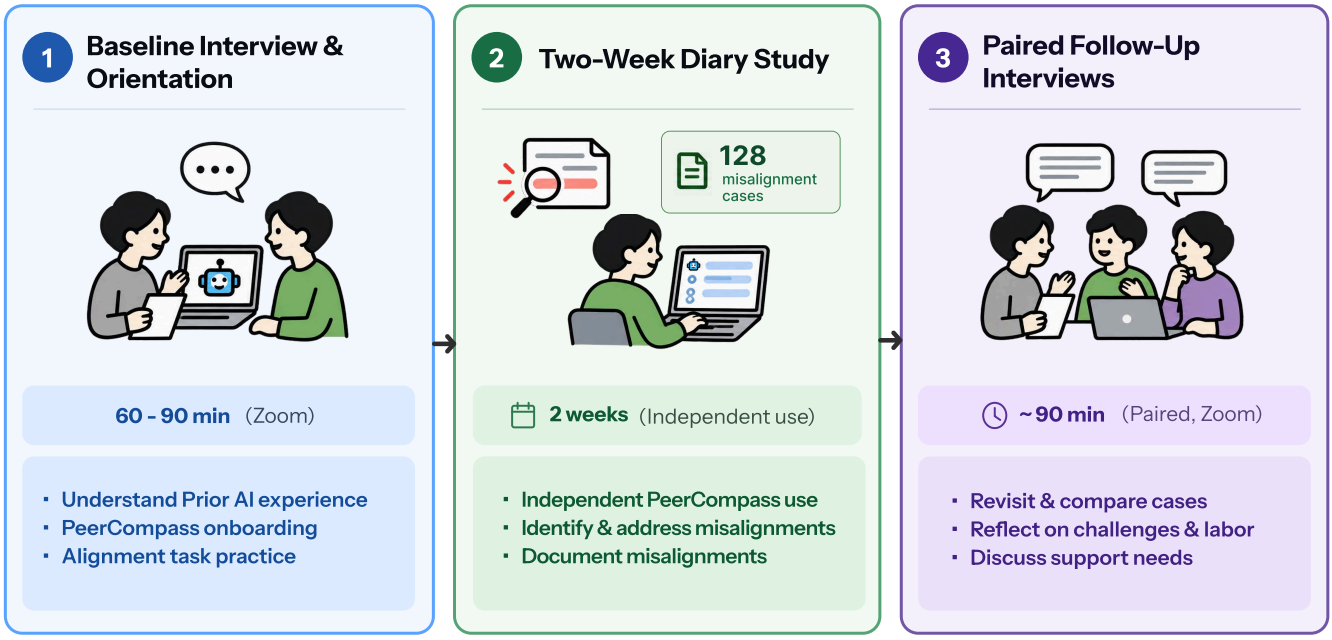


Figure 1: Overview of the three-phase study procedure, consisting of a baseline interview and orientation, a two-week diary study documenting 128 cases of misalignment, and paired follow-up interviews to revisit and compare participants' alignment judgments, challenges, and experiences.

attempt to exhaustively encode what appropriate peer support requires in every situation. This configuration allowed us to examine what alignment work arose when a system had already been given substantive domain-specific guidance. Figure 2 shows an excerpt of the system prompt alongside the participant-facing interface; the full system prompt is provided in Appendix A.

3.2 Participants

We recruited 18 certified peer specialists through social media outreach, such as LinkedIn and Facebook, and snowball sampling. Participants were eligible if they were 18 years or older, held an active peer specialist certification, and were currently or recently working with service users. They ranged in age from 25 to 56 and had approximately 9 months to 14 years of peer-support experience. The sample included 11 women, 6 men, and 1 participant who identified as gender nonconforming. Participants self-identified with diverse racial and ethnic backgrounds, including White/Caucasian, Black/African American, Afro-Caribbean, Hispanic, Asian, and multiethnic backgrounds. Educational backgrounds ranged from some college to master's degrees. Self-rated technology familiarity ranged from 5 to 7 and AI familiarity from 2 to 7 on 7-point scales. To reduce the risk of participant identification, we report demographic characteristics only in aggregate and omit individual-level age, gender, race/ethnicity, and education from Table 1. Participants received $20 for the baseline session, $10 per completed diary entry (up to $70), and $30 for the follow-up interview.

3.3 Study Procedure

3.3.1 Pilot testing. Prior to the main study, we conducted pilot sessions with two researchers experienced in community behavioral health, peer-support research, and responsible AI. The pilots assessed whether onboarding materials clearly communicated PeerCompass's intended role, misalignment examples were understandable, and diary activities were intuitive, while identifying potential risks given the sensitive nature of peer support. Based on feedback, we simplified the onboarding and diary materials, reordered activities for clarity, and introduced safeguards to reduce the possibility that prompts or examples might inadvertently distress participants who draw on their own behavioral health experiences in their work.

3.3.2 Phase 1: Baseline Interview and Orientation. Each participant completed an individual 60–90 minute Zoom session. We first asked about their peer-support practice and prior AI experiences, including tools used, tasks, and concerns. We then onboarded participants to PeerCompass by demonstrating *Chat + Explore* and explaining its intended role and functionality. Finally, we reviewed the peer-support values encoded in the system prompt and introduced the diary activity. Participants first examined researcher-prepared examples of AI responses conflicting with these values and identified and explained the misalignments they observed. They then used PeerCompass themselves and completed a practice diary entry based on a misalignment they identified. These scaffolded examples familiarized participants with identifying and articulating misalignment rather than prescribing what should count as misaligned; during the diary phase, participants were instructed to rely on their own judgment and peer-support experience. Because service users

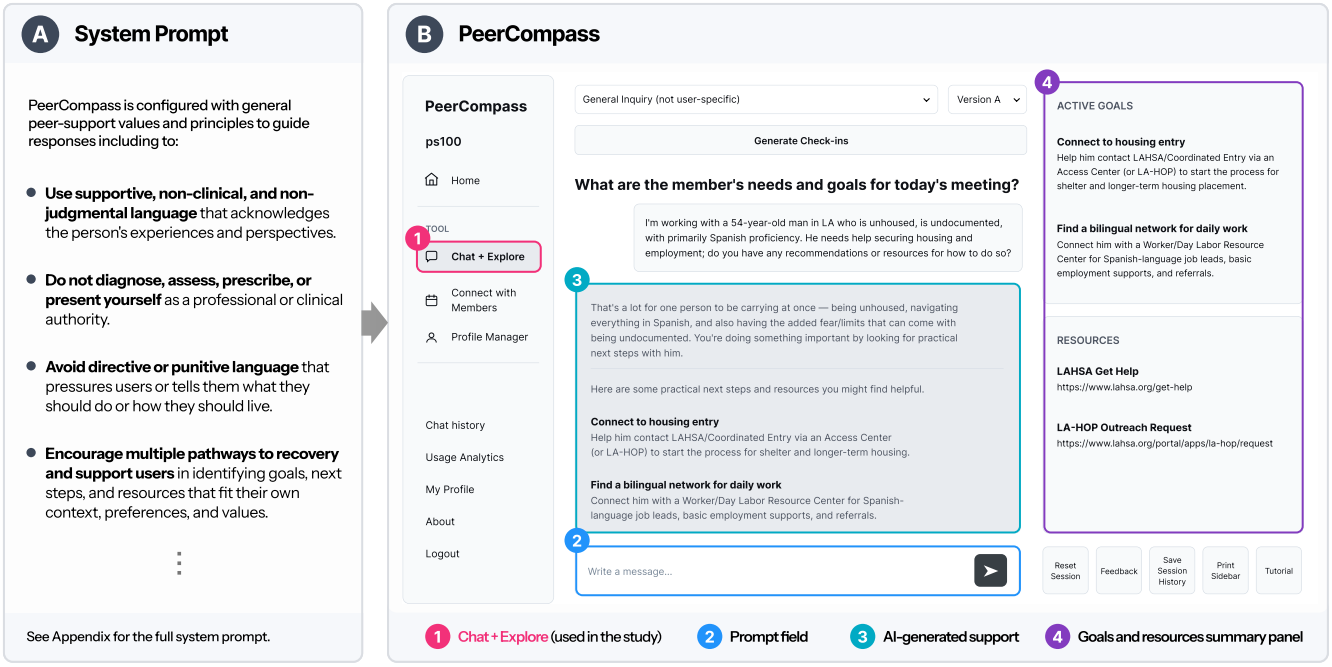


Figure 2: PeerCompass system configuration and participant-facing interface used in the study. (A) PeerCompass was configured with a system prompt encoding well-established peer-support values and principles; an excerpt is shown, with the full prompt provided in Appendix A. (B) The PeerCompass interface, annotated to show (1) Chat + Explore, the feature used in the study; (2) the prompt field, where participants could enter information about a service user's situation; (3) AI-generated support based on the information provided; and (4) the goals and resources summary panel.

had not consented to AI involvement, participants were instructed not to use PeerCompass during direct interactions with them. Instead, they used de-identified situations and experiences from their peer-support practice, excluding names and personally identifying information. This grounded system use in realistic situations without directly involving service users or entering identifiable information.

3.3.3 Phase 2: Diary Study. Participants used PeerCompass for two weeks and documented instances in which AI-generated responses did not align with peer-support values, language, or peer-to-peer practice. We asked them to intentionally explore situations drawn from their practice so they could encounter a range of potential misalignments rather than use the system only when a particular task arose. For each case, participants documented their prompt, the response they considered misaligned, why they judged it inappropriate, and the peer-support values, knowledge, or considerations informing that judgment. They then recorded how they attempted to address the misalignment and reflected on why they chose that approach and whether it addressed the misalignment. Participants could also optionally document well-aligned responses as contrast cases. Entries were submitted through Google Docs, with periodic reminders and check-ins as needed. Participants documented 128 misalignment cases, with 5–10 cases each ($M = 7.11$).

3.3.4 Phase 3: Paired Follow-up Interviews. We conducted nine 90-minute paired follow-up interviews with two participants each. Using diary entries as prompts, participants revisited selected cases and elaborated on why they considered responses misaligned, the values, knowledge, and contextual considerations informing their judgments, how they attempted to address the misalignment, and whether and why their interventions improved the support. Participants also compared possible responses to similar cases and discussed aspects of alignment they found difficult or uncertain. We concluded by asking what they learned, what they found challenging, what additional labor alignment could introduce into peer-support work, and what individual or organizational support might help. The study protocol was approved by our institution's Institutional Review Board (IRB), and all participants provided informed consent.

3.4 Data Analysis

Our dataset comprised 128 diary cases, 18 baseline interview transcripts, and nine paired follow-up interview transcripts. Interviews were recorded with consent, transcribed, and de-identified before analysis. We analyzed data from all three phases together, using baseline interviews to contextualize participants' peer-support practice and prior AI experiences, diary entries to examine their experiences identifying and addressing misalignment during PeerCompass use, and follow-up interviews to examine participants' elaborations on selected cases and reflections on challenges and labor.

Characteristic	Summary
Peer-support experience	Mean: 5.7 yrs Range: 9 months–14 yrs
Technology familiarity	Mean: 6.3 Range: 5–7
AI familiarity	Mean: 5.6 Range: 2–7

Table 1: Aggregated characteristics of study participants (N = 18). Technology familiarity and AI familiarity were self-rated on 7-point scales (1 = not at all familiar, 7 = extremely familiar).

We used reflexive thematic analysis [11, 12]. Three researchers independently open-coded an initial subset of diary entries and interview transcripts, attending to how participants identified and interpreted misalignment, the knowledge and considerations informing their judgments, how they responded to misalignment, the difficulties they encountered, and their broader reflections on the alignment process. Researchers met regularly to compare interpretations, discuss emerging patterns, revisit excerpts, and iteratively refine a shared set of codes. Rather than treating disagreements as coding errors, we used differences in interpretation to revisit the data, clarify code boundaries, and refine our analytic understanding. As analysis progressed, we compared patterns across diary cases and interviews and grouped related codes into higher-level themes. Through successive rounds of coding and discussion, we refined these themes and their relationships and developed the process model presented in our findings. We checked themes across participants, including cases that complicated emerging patterns, and refined their boundaries accordingly. To characterize recurring forms of misalignment across the diary entries, we examined all 128 diary cases, allowing multiple codes per case; category counts were therefore not mutually exclusive. Consistent with reflexive thematic analysis [12, 39], we did not calculate inter-rater reliability because our goal was to develop interpretations through collaborative and repeated engagement with the data rather than use coding agreement as a measure of validity.

3.4.1 Positionality Statement. Our research team brought backgrounds in HCI, computer science and AI, responsible AI, and mental and behavioral health research, as well as experience studying peer support and AI use in behavioral health. These backgrounds informed our study design and interpretation. Some researchers were also involved in designing and developing PeerCompass. We recognized that this relationship to the study system introduced additional considerations for how we designed the study and interpreted the data, and that participants' awareness of it might influence their engagement with the study. We therefore explicitly framed the study around identifying where PeerCompass did not fit peer-support practice and emphasized that critical feedback and misalignment cases were valuable to the research. Researcher-prepared examples were presented as illustrative rather than prescriptive, and participants were encouraged to rely on their own peer-support experience and judgment when determining what constituted misalignment.

4 Findings

Our findings reveal the labor involved in making AI-generated support fit peer-support practice, even when peer-support values were already incorporated into the system. Peer specialists still had to judge whether responses were appropriate for particular service users' values, circumstances, and relationships, judgments that the initial value specification did not fully resolve. We describe this labor by examining how peer specialists brought together complementary forms of knowledge, diagnosed misalignment, repaired AI-generated support, and evaluated its sufficiency for intended use. Throughout, we detail the challenges they encountered in carrying out this work.

4.1 Misalignment Persists After Value Specification

During the two-week diary study, participants used PeerCompass to generate support for de-identified situations drawn from their peer-support practice, including identifying resources, considering recovery goals and next steps, and responding to situations involving emotional distress, housing, employment, substance use, and mental health. These uses allowed us to examine how AI-generated support fit the needs and circumstances of particular service users. Although PeerCompass had already been configured with established peer-support values and principles, participants continued to identify responses that did not fit particular people, situations, relationships, or intended uses.

4.1.1 Purposes of Alignment in Peer Support. Participants described three interrelated purposes for aligning AI in peer-support work. First, it helped them to advocate for individual service users, ensuring that AI-generated support reflected a particular person's needs, preferences, circumstances, and lived realities. As P4 explained, *"I am the one speaking with the client, and not the AI. I have more information about this person more than the AI does. I know the best way to communicate the message to the person, so it might be tailoring the response just to suit what the person needs."* Second, they saw alignment as a way to preserve peer-support relationships, including service and trust. P13 explained, *"I was trying to protect the relationship between the peer specialist and the service user. I wanted the response to really communicate that sense of 'I'm here with you' instead of 'here's how we should handle this,' or 'I'm trying to fix someone.'"* Third, participants viewed alignment as a way of protecting peer-support practice and role boundaries. P8 noted that

"it's very easy to slip into clinical," while P9 emphasized the importance of *"staying in your lane."* For participants, alignment was therefore a way of preserving the distinctive role of peer support as AI became involved in their work.

4.1.2 Forms of Misalignment in the Diary Study. With these purposes in mind, participants evaluated PeerCompass's responses and documented 128 cases of misalignment during the diary study. Because a single case could contain multiple forms of misalignment, the following categories were not mutually exclusive. See Appendix B for an example showing how participants documented and responded to a misalignment in a diary entry.

Across these cases, recurring misalignments included incorrect or incomplete understandings of service users, their circumstances, and available resources (n=45); premature shifts toward advice, planning, or problem solving (n=54); otherwise useful guidance that crossed peer-role boundaries (n=31); communication or guidance that was difficult to enact within peer-support interactions, including responses that felt clinical, scripted, overwhelming, or otherwise relationally inappropriate (n=79); and seemingly helpful support goals or recommendations that conflicted with what the particular situation or longer-term recovery required (n=41). For example, an incomplete understanding of a service user's circumstances and available resources appeared when PeerCompass recommended a housing resource that could be helpful for some people but was inappropriate for someone needing more immediate housing support. Drawing on both having navigated the resource while homeless and supporting others through it, P1 recognized that its waitlist could extend from one to eight years. A premature shift toward advice appeared in another diary, where PeerCompass offered strengths exercises and next steps after a service user lost a job held for nearly twenty years. P16 judged that these could be helpful later, but that the person first needed space to experience the accompanying grief and loss of identity.

Importantly, these misalignments did not necessarily take the form of direct violations of the peer-support values incorporated into PeerCompass. Responses could appear broadly consistent with those values while still failing to fit a particular service user, situation, relationship, or intended use. The 128 cases therefore show that alignment work could remain after value specification—not simply because specified values were violated, but because peer specialists still had to determine what appropriate support required in particular cases.

The following sections unpack this remaining alignment work: the complementary forms of knowledge peer specialists drew on, how they diagnosed and repaired misalignment, how they judged whether the resulting support was sufficient for intended use, and the challenges that arose at each stage.

4.2 Combining Complementary Forms of Knowledge for Alignment

To align AI-generated support with peer-support practice, participants brought together three complementary forms of knowledge: (i) their understanding of particular service users, which helped them anticipate how support would be received; (ii) their lived experience, which highlighted practical and emotional realities; and (iii) their practice knowledge, which supplied principles and patterns for assessing support.

4.2.1 Drawing on Knowledge of Service Users to Anticipate Reception. Because a response that felt supportive to one person could overwhelm another, participants drew on their knowledge of particular service users and relationships to judge how AI-generated support might be received. P12 explained that a relationship could provide *"a hint about their personality traits, on how they perceive things, on some of their views,"* while acknowledging that responses remained uncertain. P6 similarly explained, *"I know someone who I'm talking to, if I give them this response, they will be able to handle it, but then others, they definitely will not be able to handle it."* Such knowledge helped participants assess whether otherwise plausible support suited a particular recipient. At the same time, knowing a service user more deeply could make alignment more demanding rather than easier. P17 contrasted interactions with people they had just met with supporting someone they had known for years: *"you have to factor in a lot of different things since you know them,"* because *"there's more that you could lose or harm the relationship if you don't have the right response."*

4.2.2 Drawing on Lived Experience to Recognize Realities AI May Overlook. To assess whether seemingly appropriate AI guidance accounted for the realities of receiving support, participants drew on their own experiences of vulnerability, recovery, and practical constraints. P1, for example, questioned seemingly appropriate guidance for people experiencing homelessness based on their personal experience of having been homeless. In one diary entry, PeerCompass suggested calling Nevada 211 to a person experiencing homelessness. They noted that this recommendation overlooked a basic constraint: many people experiencing homelessness may not have access to a phone. Drawing explicitly on their own experience, they wrote, *"I firsthand understand this response is inappropriate because I experienced homelessness myself and struggled to find a phone to use during that time."* In the follow-up interview, they similarly recalled, *"I remember not having a phone [...] just saying to use the phone if you're homeless may not be appropriate,"* describing this judgment as based *"mostly off of my own personal lived experience."* Together, the diary entry and interview account show how their lived experience enabled them to recognize a practical constraint that was not apparent from the resource recommendation itself. P6 similarly drew on experience *"as a person who has needed this peer support,"* considering how they would have wanted to be assisted and what language might have felt triggering. Lived experience thus provided an experiential reference point for judging whether AI-generated content fit the realities of receiving support.

4.2.3 Drawing on Practice Knowledge to Interpret Appropriate Support. To interpret what peer-support values required in a particular situation, participants drew on training, accumulated practice experience, and knowledge shared by other peer specialists. P5 described drawing on both *"my training as a certified peer specialist"* and *"my day-to-day experience working with service users."* They also drew on established patterns for interpreting what support required in particular situations. P17 attributed confidence in distinguishing *"what I should do and what I shouldn't do"* to years of peer-support practice. P18 similarly emphasized that, alongside formal resources

and training, *"at the end of the day [...] it's about hands-on experience,"* including *"learning the hard way and figuring out on your own."* This knowledge helped participants judge whether AI's proposed support matched the kind of engagement a person needed at that moment.

4.3 Stage 1: Diagnosing AI's Fit for Peer Support

Diagnosing misalignment required peer specialists to determine where and why AI-generated support failed to fit peer-support practice. These judgments spanned five dimensions of peer fit and involved several challenges, including subtle mismatches, uncertainty, and competing peer-support priorities.

4.3.1 Diagnosing Misalignment Across Five Dimensions of Peer Fit. Diagnosing misalignment required participants to examine how AI-generated support would function for a particular person, within a peer relationship, and at a particular moment, while also considering role boundaries and longer-term recovery. These judgments involved looking beyond a response's apparent usefulness. Support that seems reasonable in one respect can become inappropriate in another.

Assessing contextual appropriateness. Participants examined the assumptions behind AI recommendations and considered how seemingly reasonable guidance would function in a service user's circumstances. When PeerCompass recommended a detox center as a strong starting point for an unhoused service user actively using fentanyl, P1 recognized that the center required a referral and could involve waiting days for a bed. Because *"acting in the moment is crucial"* when someone is ready to pursue sobriety, P1 judged that *"asking them to wait for days [...] isn't feasible."* Similarly, when a service user burned objects as a protective ritual tied to a delusional belief, PeerCompass attempted to reduce harm by suggesting safer substitutes, such as writing or holding an object. P12 recognized that offering an alternative "protective" ritual could *"sound like you are encouraging or participating in the delusion."* These assessments required looking beyond a recommendation's apparent helpfulness to anticipate whether and how it would work in the person's actual circumstances.

Assessing relational appropriateness. Participants also considered whether AI-generated support would sustain authentic and emotionally supportive peer interactions. P12, for instance, found a question about what was drawing a young adult toward education *"insensitive and out of touch,"* reasoning that it could undermine rapport and create *"a disconnect between the peer and peer specialist."* Participants also considered how communication would function in actual interactions. Drawing on experience with aggressive service users, P11 worried that AI's suggested tone could *"come off as patronizing"* and that AI was *"underestimating how angry some service users can get"* and *"how quickly their anger can get out of control."* Across these judgments, relational appropriateness concerned whether AI support remained emotionally attuned and responsive to the dynamics of real peer interactions.

Assessing temporal appropriateness. Relational fit also depended on when support was offered. Participants distinguished potentially helpful guidance from guidance appropriate at the current moment. P13 identified responses that *"shifted too quickly into problem solving,"* explaining, *"If someone has told me something really vulnerable, I want them to feel heard before we think about what comes next."* P3 similarly noted that people were often more receptive to safety options *"after they feel fully heard."* Participants thus had to assess readiness and sequence as well as the content of support.

Assessing peer-role appropriateness. Participants further judged whether guidance was appropriate for them to deliver as peer specialists. P15 explained, *"Sometimes it can be useful information, but still not be appropriate for me to deliver as a peer worker. I need to recognize where my role ends."* P12 likewise described recommendations as *"accurate and helpful"* yet *"too directive, clinical and therapeutic."* These judgments required distinguishing the potential usefulness of information from its fit with peer responsibilities and boundaries.

Assessing recovery appropriateness. Participants considered how responses might support longer-term recovery while respecting immediate concerns and preferences. P1 challenged AI's accommodation of a service user's reluctance to engage with clinical care following prior negative experiences. While respecting *"power of choice,"* P1 emphasized, *"We can't avoid clinicians. Peer support is not a replacement for a clinical role. Our job is to help facilitate that person get to a clinician, if that's something that they need."* P8 also considered whether AI sustained hope in recovery itself: *"Reminding the peer that recovery IS possible is key."* Recovery appropriateness thus involved considering whether AI support sustained hope and supported personally meaningful pathways toward recovery while respecting service users' autonomy and preferences.

4.3.2 Challenges in Diagnosing Misalignment. Participants' labor in this stage involved scrutinizing increasingly subtle mismatches, determining what counted as misaligned under uncertainty, and weighing competing peer-support priorities. These demands persisted even when responses appeared broadly useful and consistent with peer-support values.

Increasingly subtle judgments of peer-support fit with limited visibility. As AI produced seemingly supportive responses, peer specialists still had to judge whether subtle features of that support fit their role and relationships with service users. This included judging when empathy shifted into directing the person, whether advice could feel patronizing, or whether problem solving came before the person felt heard. P9 contrasted earlier stages of repair, when problems were easier to identify, with later stages in which it could take *"hours and hours and hours to identify subtle nuances."* They anticipated this difficulty continuing as AI became more capable: *"as AI is getting stronger, more frustrating for the worker the further you go along in it."* Yet the effort involved in these judgments was not necessarily reflected in the amount of content participants contributed. As P9 explained, *"you're not giving them content like you did in the beginning [...] So it's invisible because you're spending so much more brain power and time."*

What could remain hidden, then, was the work of determining whether an apparently empathetic response would enact peer support appropriately for a particular person and relationship. A small change in wording or sequence could reflect considerable deliberation about preserving choice, maintaining connection, and staying within peer-role boundaries. This gap between evaluative

effort and visible changes also raised concerns about organizational recognition. P15 argued that routine AI repair should be *"recognized as actual work rather than something we just expect to do on top of everything else,"* calling for *"protected time, training, and clear expectation."* These accounts highlight how the judgment required to make AI-generated support usable in peer practice could be difficult to recognize and resource when its visible result was only a modest adjustment to the response.

Determining the boundary of misalignment under uncertainty. Participants sometimes found it difficult to disentangle useful support from subtle misalignment and establish confidence in their judgments. P5 explained, *"The most challenging part wasn't really identifying the responses that were completely wrong, it was identifying the ones that were possibly good and partly misaligned."* A response could begin with empathy before shifting toward extensive advice, leaving P5 to work out *"exactly where the response stopped feeling like peer support and started feeling more like counseling or case management."* This difficulty was compounded by recognizing that other peer specialists might reasonably disagree. For P6, these ambiguities made it difficult to trust their own assessments: *"Especially the things I've said in the gray area, you aren't really sure where these ones are going to fit. I have doubted my judgment a lot."* They wanted *"the opinion of another peer specialist"* to help them judge these uncertain cases.

Anticipating the recipient's response introduced a further difficulty. P13 explained, *"None of us can exactly know how another person is going to receive a response [...] what feels supportive to one person might not really feel the same to someone else."* Although experience meant they were *"not really guessing completely"*, it did not resolve these doubts. Participants thus faced the demanding work of deciding where support crossed into misalignment while questioning their own interpretations and lacking certainty about how others would judge or receive it.

Navigating competing peer-support priorities. Diagnosis became difficult when AI addressed multiple legitimate peer-support priorities but balanced them differently than participants judged appropriate. In one diary entry, P7 considered a service user who described withdrawing after a friend's death as needing space to grieve, while providers worried about psychiatric deterioration. PeerCompass acknowledged both grief and safety, but then moved into monitoring sleep, medication, self-harm, and other "red flags," along with safety planning. P7 did not consider safety concerns inappropriate, but judged them premature given the absence of an immediate risk, potentially turning grief into something to be *"observed, assessed, and managed."* Diagnosis therefore required judging not simply which priorities mattered, but how much weight they should receive and when in the particular situation.

4.4 Stage 2: Repairing AI for Peer Support

Once participants identified a misalignment, they worked to make AI-generated support more appropriate for peer-support practice. They did so through five repair strategies while navigating challenges in translating practice knowledge, managing trade-offs, and sustaining repairs across iterations.

4.4.1 Reworking AI-Generated Support Through Five Repair Strategies. Identifying misalignment left participants with the work of developing an appropriate alternative. They had to determine what to retain, redirect, or replace, and translate these decisions into changes through re-prompting, direct editing, selective curation, or their own contributions. These repair strategies did not map one-to-one onto particular forms of misalignment; participants drew on different strategies, sometimes in combination, depending on what needed to change to make the support appropriate. Making support usable could require changing its objective, correcting gaps or assumptions in AI's understanding, reorganizing its sequence, or reworking how it could be enacted and communicated.

Reorienting AI's support goals. Participants had to redefine what AI should help accomplish when its proposed objective did not fit peer-support practice. When AI immediately constructed an employment plan, P7 re-prompted it to emphasize *"open-ended questions, self-determination, mutuality, and following the person's own goals"* and first explore *"what work meant to him, what kind of work he wanted, and what support he would find useful."* This redirected support toward understanding the person's aspirations before developing a plan. A diary entry from P14 further illustrated how reorientation could involve changing whose behavior the support targeted. When PeerCompass focused on helping a service user at risk of losing shelter housing adapt her behavior to the program, P14 objected that *"most of the responses were about how she should react with the team vs. how the team should align themselves,"* with *"very little peer equity."* Rather than centering compliance with the shelter, P14 proposed zooming out to the service user's longer-term goals and creating opportunities to revisit program expectations while preserving *"voice and choice."* In another case, P1 redirected AI's response to a service user's reluctance toward therapy: *"Do not give her options that do not include clinicians based on her resistance to therapy, but methods on how we can get her there."* These repairs involved reconsidering what support should accomplish and whose goals it should center.

Correcting gaps and assumptions in AI's understanding. Participants had to explain circumstances and anticipated consequences that AI had overlooked, then specify how its guidance should change. When PeerCompass suggested storing a recovery plan on a phone, P10 noted, *"Not all consumers have phones,"* and adapted the suggestion to an index card that could be kept in a wallet. In another diary entry, they challenged the AI's characterization of reconnecting with a service user's father as one of his *"strongest motivators,"* explaining that a previously expressed interest did not make it a current priority; rather than acting on the suggestion, they reserved it for if the service user raised the goal himself. P6 similarly explained that imposing a narrow timeframe for seeking help could *"create feelings of panic or anxiety"* and redirected AI to remove deadline-related language. This work involved identifying what AI had misunderstood and translating that understanding into a concrete adjustment to support.

Reordering the sequence of support. Even when guidance was potentially useful, participants had to reorganize it around emotional readiness and practical prerequisites. For example, P16 described a man who felt useless after losing a job he had held for nearly twenty years. PeerCompass responded with reflective questions, a strengths inventory, and practical next steps. P16 found these suggestions potentially useful but premature, describing them as *"less like peer support and more like career coaching."* Across

several re-prompts, P16 progressively removed questions, strengths exercises, and action steps as the AI continued to move toward rebuilding confidence, ultimately asking it to focus on *"listening and emotional support"* without coaching. P16 judged the result successful because it kept the person's grief and loss of identity at the center and allowed them to move *"at their own pace."* Repair thus deferred potentially useful guidance until the relational moment made it appropriate. Before recommending virtual care, P1 similarly instructed peer specialists to *"assess their computer skills first, get that addressed and then move forward."* Participants thus had to determine what should happen first, what should wait, and what would make later guidance usable.

Translating AI guidance into actionable peer practice. Participants also had to supply practical steps when AI offered abstract ideas without explaining how to use them in peer interactions. In one diary entry, PeerCompass suggested making a service user's goal *"identity-based,"* but P10 could not envision how to enact the idea in a session, finding no *"clear action step."* P10 translated it into a concrete question—*"What do you find that is already helpful for motivating you to accomplish things?"*—which led the AI to generate more usable, strength-based questions. P8 similarly emphasized translating information into *"language that would be easy for anyone to understand"* and that they would be *"comfortable using with someone."* This repair required working out how a general recommendation could become a concrete question or action that a peer specialist could explain and use.

Reconstructing AI's communication for peer relationships. Participants had to reshape how AI-generated content was expressed so that useful guidance could function within a natural peer conversation. In one diary entry, P2 asked how to support a college student experiencing panic attacks and falling behind in classes. PeerCompass offered coping strategies, campus supports, and scripts for contacting professors. Although P2 found much of the information useful, they felt its delivery positioned the peer as directing rather than speaking alongside the service user. P2 therefore retained useful information while removing scripted language, reducing the amount of content, and reframing recommendations as options the student could take or leave. P2 judged the result more appropriate because it felt *"more like sitting next to someone instead of directing them."* P12 similarly found an otherwise useful response *"formulaic and like therapy,"* prompting AI to *"sound like a natural free flowing conversation,"* while P10 chose to *"not include the scripted response"* and use their own language for delivery. Repair thus involved translating useful AI content into a communicative form that was conversational rather than scripted, suggestive rather than directive, and preserved the service user's voice and agency.

Across these strategies, repair was therefore not simply a matter of editing problematic AI content. Participants reworked what support was trying to accomplish, corrected what it assumed about the person, determined when otherwise useful guidance should enter the interaction, translated abstract guidance into something they could enact, and reshaped the relational position from which support was offered.

4.4.2 Challenges in Repairing Misalignment. Repair required peer specialists to bridge AI and peer-support practice, manage tensions introduced through repair, coordinate alignment work with relational attention, and maintain accumulated repair knowledge across iterations.

Developing expertise to bridge AI and peer support. Knowing what appropriate peer support required did not automatically equip participants to make AI produce it. Repair required them to learn how to instruct AI and adapt its outputs while taking on teaching, knowledge-providing, editing, and training roles (P4). P3 described translating peer-support concepts into *"specific feedbacks that the AI could use to improve its response."* Making practice knowledge explicit was particularly difficult when conveying a person's circumstances and relationships: P6 struggled with *"how to incorporate all these without even telling the life story of this person."* Translation also ran in the other direction, as P9 combined information about the service user with AI-generated content to make responses more empathetic. Participants had to develop these capabilities through repair itself. P15 described *"learning by doing,"* explaining, *"I didn't have a formal course that taught me exactly how to translate peer support value into prompt."* This reliance on individual learning prompted requests for collective support, such as P13's proposal for regular discussions to share responses and reasons for revisions. Repair thus demanded that peer specialists expand their roles and knowledge while largely working out for themselves how to bridge AI and peer-support practice.

Repairing one dimension without creating misalignment in another. Participants could not assume that improving one aspect of support would make the overall response more appropriate. P13 described *"lots of times"* when *"the first change improved one issue, but introduced another issue"*; attempts to increase empathy could produce too many reflective questions or still move too quickly toward problem solving. Participants therefore had to inspect the effects of each intervention and adjust again when a repair disrupted another aspect of peer support. Providing more context introduced a related difficulty: participants needed useful, personalized responses while limiting disclosure about service users. P9 described this tension as *"how much do you want to not identify so that you get helpful answers."* Repair could also create tension with the interaction itself when carried out during peer sessions. P9 worried that working with AI could mean *"I'd be looking at AI and not eye to eye with the service user. So breaking a personal human connection,"* while P18 cautioned that attending to AI could *"take away from the person-to-person connection."* Repair therefore required coordinating not only the effects of changes within the output, but also the demands of alignment work with the relational attention needed for peer support.

Managing repair across iterations. Repeated repair required participants to keep track of earlier interventions, their effects, and unresolved mismatches to decide what to try next. P7 recalled *"scrolling through the prompts and trying to remember exactly what had been changed,"* while P15 described adjusting their approach based on *"what was still misaligned."* Earlier instructions also did not reliably carry through into subsequent responses, requiring participants to reinforce the same expectations. P13 explained, *"I wasn't just fixing one response. I was repeatedly teaching it to stay with emotions longer."* Participants thus had to reconstruct the history of their repairs and repeatedly restate peer-support expectations to sustain progress across iterations.

4.5 Stage 3: Judging Sufficiency for Use

Repair did not necessarily continue until AI produced a perfectly aligned response. Instead, peer specialists judged whether the response was *sufficiently aligned for its intended use* and whether remaining misalignments warranted further intervention. These judgments involved three considerations: the consequentiality of residual misalignment, how the output would be used in practice, and the marginal value of further AI repair relative to the effort it required. Thus, sufficiency was not a fixed threshold of alignment, but a situated judgment about whether an output was aligned enough for how it would actually be used.

4.5.1 Judging Sufficiency Based on Remaining Misalignment, Intended Use, and the Value of Further Repair. Stopping AI repair did not require a perfectly aligned response. Participants had to weigh the significance of remaining mismatches, the intended use of the output, and the likely benefit of further repair relative to its effort.

Assessing the significance of remaining misalignment. Participants had to examine what remained misaligned and judge how consequential it was for peer support. P15 described deciding whether a *"meaningful peer support issue"* still remained, such as a response that was still too clinical, directive, or disconnected from the person's experience, and whether the resulting support would leave the service user feeling *"heard, respected and emotionally safe."* In their diary, after two rounds of re-prompting produced a more emotionally supportive response, P15 stopped because the remaining differences were *"mostly about wording"* that they could adjust themselves in conversation. The judgment therefore concerned whether a remaining mismatch warranted further intervention rather than merely further stylistic refinement. P13 similarly stopped when *"additional changes would have been more about my personal style rather than improving its alignment with peer support values."*

Evaluating fit for the intended use. Participants also considered how they would use the output and what adaptation that use would entail. P9 reflected, *"The AI was never meant to copy and paste for the end user, so clinical wording wouldn't matter as much as long as [the] peer understood it."* This orientation also appeared in their diary, where P9 treated AI responses as *"helpful and good reminders of directions a peer specialist could take with a peer."* Clinical wording could therefore be acceptable in material that a peer specialist would interpret and adapt, even if it would be inappropriate to deliver directly. P13 similarly stopped when a response *"felt like something I could actually see in a peer conversation,"* explaining, *"I didn't really feel the need to make it perfect because in practice I will continue adapting my words based on how the service user responded."* Assessing readiness required anticipating how the output would enter peer practice and which adjustments participants would continue making themselves.

Determining whether further AI repair is worthwhile. Participants judged whether another attempt at prompting was likely to improve the response enough to justify the effort. P11 stopped when the remaining problems reflected AI's *"overall assumptions and approach"*, explaining, *"I felt I would spend more time trying to guide the AI than I would responding based on my own professional judgment."* This decision required comparing the likely benefit of further prompting with the work of developing appropriate support themselves. Stopping AI repair could thus mean taking over unresolved alignment work, even when the response remained unsuitable.

4.5.2 Challenges in Judging Sufficiency for Use. Readiness judgments were complicated by the lack of a stable reference point for sufficient alignment and the need to sustain critical judgment across repeated evaluation.

Lacking a stable reference point for sufficient alignment. Participants lacked a clear reference point for determining when an AI response had become sufficiently aligned. Improvement over a previous response did not necessarily mean that it was ready for practice, while further refinement was almost always possible. P15 described this as *"an endless process where there is always another word or phrase you could change,"* while also questioning *"whether the changes really addressed the problem, or whether the AI had just changed the wording while keeping the same underlying issues."* Nor did differences from their preferred wording necessarily indicate misalignment. Thus, being better than the previous response was not sufficient evidence of readiness; participants had to continually anchor their stopping judgment in whether meaningful misalignment with peer-support practice remained.

Sustaining critical scrutiny after repeated repair. Participants had to continue evaluating responses even after investing effort in correcting them. Incorporating their feedback did not itself establish that a response was appropriate; each revision still required scrutiny against the demands of peer-support practice. P17 emphasized the need to maintain critical scrutiny across repeated repair, judging what remained appropriate *"regardless of how many times I prompted the AI and how many different responses there were."* This placed an ongoing demand on participants to exercise their own judgment throughout repair, including when deciding that enough had been done. Participants attributed this capacity partly to accumulated practice experience and training, and questioned whether newer peer specialists would be equally equipped to undertake it. P18 warned that a newer peer specialist relying on AI might not yet have the *"critical thinking judgment"* needed to recognize when a revised response remained inappropriate. Repeated repair therefore continued to demand peer-support expertise at the point of acceptance, leaving peer specialists responsible for assessing whether their interventions had actually produced usable support.

5 Discussion

5.1 The Last-Mile Labor of Alignment

Our findings illustrate that substantial alignment work persisted even when relevant practice values had already been incorporated into the AI system. This work arose not only when AI failed to follow specified values, but also when responses appeared broadly consistent with them yet did not fit a particular person, relationship, or situation. Peer specialists therefore still had to determine what appropriate support required in each case and adapt AI-generated support accordingly. In this sense, AI did not eliminate the human work required to produce appropriate support, but redistributed it: as the system took on content generation, peer specialists took on the work of evaluating and adapting that content for particular service users and situations.

This finding extends research showing that automation can redistribute rather than eliminate human labor [20, 30, 48, 63]. Drawing on Gray and Suri's metaphor of the *last mile of automation* [23], we conceptualize this work as *last-mile alignment labor*: the human effort required to bridge what AI produces and what is appropriate for particular people and situations in practice. As AI took on the visible production of support, peer specialists remained responsible for evaluating whether that support fit, repairing it when necessary, and determining when it was sufficiently appropriate for use. Our findings reveal three shifts through which this labor was redistributed and often made less visible.

From visible content production to less visible evaluative work. When peer specialists produced support content directly, their contribution was reflected in what they created. With AI generating candidate content, much of their contribution instead lay in evaluating and revising outputs in ways that were not readily visible in the finished response. Even accepting a suggested resource as appropriate could require substantial interpretive effort without producing a visible edit. A polished output could therefore appear useful on its own, obscuring the human judgment needed to establish its suitability and leaving that evaluative contribution invisible.

From applying domain knowledge directly to translating between AI and service users. Before using AI, peer specialists could draw on their knowledge directly to formulate support for service users. With AI generating candidate support, that knowledge also had to be translated through the system. Participants translated their understanding of a particular service user's values, circumstances, and support needs into instructions that could guide AI behavior, and then interpreted and adapted AI-generated content back into support appropriate for that person. Knowing what appropriate peer support required therefore did not automatically translate into knowing how to make AI produce it. Participants learned through trial and error how to bridge these two forms of work, but this translation effort was not readily visible in the finished response.

From output completion to ongoing coordination of repair and use. A completed AI response did not mean that support was ready to use, nor did repair necessarily continue until the response was perfectly aligned. Peer specialists had to decide whether remaining mismatches warranted further prompting, could be addressed through their own adaptation, or made the output unsuitable for use. These decisions depended on how the output would be used and whether further AI repair was worth the effort. Participants also anticipated having to coordinate this evaluative work with listening and maintaining connection during peer interactions. These decisions about where to direct effort and attention were not readily visible in the completed output.

Together, these findings suggest a redistribution of work when AI is incorporated into peer support, shifting effort from producing content directly toward evaluating outputs, translating between AI and service users, and coordinating repair and use—forms of work that are less visible.

5.2 How Last-Mile Alignment Labor Unfolds and Why It Persists

The redistribution described above raises a further question: what does this last-mile alignment labor consist of, and why does it persist? Prior HCI research has shown how people adapt general-purpose AI to the values and requirements of particular domains and practices [3, 25, 38, 81]. Our findings extend this work by characterizing the alignment work that remains after relevant practice values have already been incorporated into the system, particularly when peer specialists align AI-generated support for others. Unlike alignment for one's own use, peer specialists were aligning AI-generated support on behalf of another person who did not interact with the system directly. This required them to interpret what broad practice values demanded for that person's circumstances and relationship while remaining responsible for how the resulting support would be used.

We synthesize this remaining work as an iterative process of diagnosing misalignment, repairing AI-generated support, and judging its sufficiency for use (Figure 3). This process was grounded throughout in complementary forms of knowledge. Participants drew on accumulated practice knowledge, lived experience, and their understanding of service users and relationships to identify where and why AI-generated support failed to fit. They then repaired its assumptions, goals, sequencing, practical steps, or language. To judge sufficiency, they weighed the significance of remaining mismatches, how the output would be used, and whether the likely benefit of further AI repair justified the effort it required. Unresolved concerns or new mismatches introduced through repair could prompt a return to diagnosis and revision. Alignment therefore did not proceed toward a fixed endpoint, but through repeated judgments about what needed to change and when the resulting support was sufficient for a particular use.

Our findings also help explain why this labor persisted despite prior value specification. Broad peer-support values provided direction without fully determining what appropriate support required in a particular case. For example, AI guidance on how to carry out a potentially harmful burning ritual more safely could appear consistent with harm reduction, yet when that ritual was tied to a service user's delusional belief, the same guidance could risk reinforcing that belief. Peer specialists therefore had to judge not only whether AI-generated support reflected relevant values, but what those values required when enacted for a particular person, relationship, situation, and intended use.

This persistence resonates with longstanding accounts of *situated action*, which challenge the assumption that general plans and representations can fully determine action in advance [8, 56, 57, 68]. Suchman [68] argues that plans can orient action, but their relevance must be worked out in relation to the contingencies of situations as they unfold. We find a similar relationship between value specification and alignment in use: specifying relevant values can orient AI behavior without eliminating the situated judgment required to determine what appropriate behavior means in a concrete case. Our findings do not suggest that better prompts, models, or domain-specific alignment could not reduce this work. Rather, even with explicit, literature-grounded peer-support values, participants still encountered cases requiring situated judgment about

particular people, relationships, situations, and intended uses. This suggests that value specification and support for situated human judgment are complementary design goals.

5.3 Design Implications for Supporting Alignment Work in Peer Support

Our findings identify opportunities to support last-mile alignment: the judgments, learning, and coordination required to make AI-generated support usable. Building on proposals for user participation in alignment [3], we outline support mechanisms that address challenges across diagnosis, repair, and judgments of sufficiency.

Diagnosing: Support uncertain judgments and make diagnostic reasoning visible. First, systems could help peer specialists keep multiple dimensions of peer support in view when diagnosing misalignment. Difficult cases often involved legitimate considerations that could not be straightforwardly satisfied at once, such as emotional validation and timely safety intervention or respect for autonomy and encouragement toward recovery. Building on interfaces that externalize multiple dimensions to help people reason across competing considerations [49], we extend this approach to alignment diagnosis: systems could surface the relevant peer-support dimensions together and indicate which may be prioritized, compromised, or remain unresolved under different responses. Rather than determining the "right" balance, such mechanisms could help peer specialists manage these considerations against their situated knowledge of the service user and make the reasoning behind their diagnosis more explicit.

Second, systems could treat uncertainty itself as a legitimate diagnostic state and preserve it for later reflection. Prior work has shown the value of externalizing people's rationales so that their judgments can become objects of critical reflection [82], while uncertainty representations alone may leave people unsure of how to act on them [55]. Extending these approaches from resolving uncertainty in the moment to preserving unresolved judgments, interfaces could allow peer specialists to defer a difficult judgment while recording their concerns and reasoning. These records could later be revisited after gaining additional situated experience, alongside colleagues' perspectives, or when similar cases arise. This would allow uncertainty to become a resource for continued individual and collective reflection rather than forcing an immediate accept-or-revise decision. Over time, such records could also make otherwise invisible diagnostic effort visible [63] and help organizations identify recurring gray areas that warrant collective discussion, additional training, or system improvement.

Repairing: Manage prior improvements and support learning across AI and peer-support practice. First, systems could help peer specialists manage trade-offs introduced by repair by making its potential effects on other dimensions of peer support visible. Building on deliberation systems such as ConsiderIt, which make contrasting considerations available for comparison and reflection [33], interfaces could preview how a proposed repair might affect other dimensions, such as empathy, conversational sequencing, or role appropriateness, before it is applied. For example, increasing emotional validation might delay a transition toward problem solving. Rather than prescribing the "right" repair, such previews could help peer specialists anticipate how improving one aspect of support might introduce tension elsewhere and consciously manage these trade-offs.

Second, systems could preserve *repair context* across service user interactions. Building on mechanisms that retain and surface users' prior preferences and interaction-derived information [31, 79], systems could preserve repair principles that peer specialists repeatedly apply for particular service users or types of situations and surface them when relevant cases arise. For example, a repair principle developed when supporting someone experiencing delusions could be retained together with the context in which it was appropriate and resurfaced in a relevant future interaction. Such persistence could reduce the repeated labor of reconstructing prior judgments or providing the same instructions while allowing repair knowledge to accumulate over time rather than disappear as a one-off intervention. Given the sensitivity of behavioral-health contexts, however, such persistence would require careful attention to consent, data minimization, and mechanisms for reviewing or removing retained context.

Third, systems and organizations could support *bidirectional learning from repair knowledge* across peer specialists. Prior work has shown that users share strategies for addressing problematic AI behavior through online communities [18]. Extending such sharing to mediated alignment for others, systems could build a shared, anonymized repertoire that connects a particular misalignment with the repair instruction used, the resulting change in AI output, and how that output was subsequently adapted in a peer conversation. Rather than serving as a library of decontextualized prompts, this repertoire could preserve the context and rationale for why a repair was appropriate. Collectively reviewing and developing these cases could reduce the burden on individual peer specialists to develop alignment strategies from scratch while supporting bidirectional learning at the organizational level: how to express peer-support knowledge in forms AI can act on, and how to translate AI-generated content into appropriate peer support.

Judging sufficiency: Support practice-grounded stopping judgments. Systems could help peer specialists retain sight of the concern that originally motivated repair. Building on Wadinambiarachchi et al. [75]'s proposed strategies for mitigating fixation by redirecting attention beyond existing AI-generated content, we extend this idea to last-mile alignment, where repeated repair can narrow attention toward the current output and away from the misalignment that prompted repair. Systems could preserve the original diagnosis as a reference point and periodically reconnect revisions to it, helping peer specialists maintain critical scrutiny across repeated repair by asking whether the original concern has actually been addressed. They could also prompt consideration of remaining misalignment and whether further revision warrants the effort. When uncertainty persists, peer specialists could defer the judgment for later review rather than continuing to refine the output indefinitely.

Beyond output alignment: Align AI use with the relational conditions of peer support. Following HCI scholarship that questions whether technological intervention is always the appropriate design outcome [9], our design implications should not be read as advocating for greater integration of LLMs into peer-support practice. The question is not only whether AI-generated support

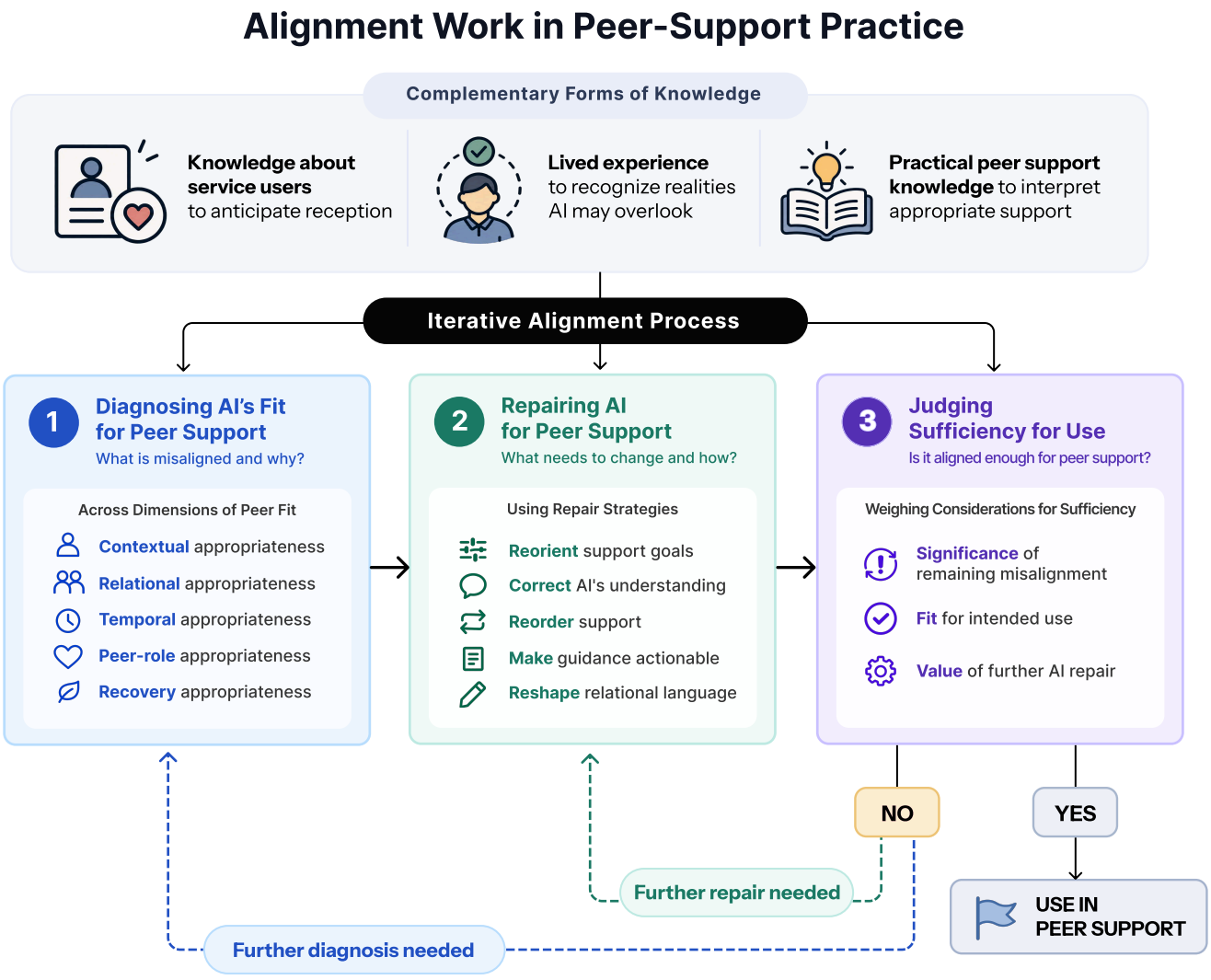


Figure 3: Process model of alignment work in peer-support practice, showing the complementary forms of knowledge that ground judgment and the iterative stages of diagnosing peer fit, repairing AI-generated support, and judging sufficiency for use in peer support.

can be made to fit peer-support practice, but whether the work required to achieve that fit can be carried out without undermining the relational conditions of the practice itself. Evaluating and repairing AI can redirect attention away from listening, responding to unfolding emotions, and sustaining connection and trust. Thus, even when alignment work successfully makes an AI output more appropriate for peer support, the work of producing that alignment may itself interfere with the relational presence through which peer support is enacted.

Alignment support should therefore help peer specialists manage when and how AI work enters peer interactions. Building on work showing that lightweight contextual cues can help people return attention to a primary task after interruptions [5, 50], systems could provide brief cues to redirect attention toward the service user when AI work begins to compete with relational presence, or support review outside of sessions altogether. Systems and organizations should also treat decisions to pause, limit, or forgo AI use in particular interactions as legitimate outcomes. The goal is not to maximize AI use or alignment activity, but to ensure that efforts to make AI fit peer support do not undermine the relational practice they are intended to support.

6 Limitations and Future Work

Our study has several limitations. First, we examined alignment labor in peer support using one LLM-based system in which peer-support values were specified through a system prompt. The forms

of knowledge, judgment, repair, and labor we observed may differ across domains, AI systems, and ways of configuring AI with practice values. Future work should examine which aspects of this alignment labor extend to other settings and technical configurations.

Second, participants worked with situations drawn from their practice rather than using PeerCompass during actual interactions with service users. Moreover, although our scaffolded examples were intended to familiarize participants with identifying misalignment rather than prescribe what counted as misaligned, explicitly orienting participants toward misalignment may have heightened their attention to it. Future studies could examine how alignment work unfolds during actual practice, including when misalignment is noticed, overlooked, left unaddressed, or addressed through adaptations outside the AI system.

Third, our participant pool may have skewed toward peer specialists who were already relatively comfortable with technology or motivated to engage with AI tools. Indeed, participants generally reported high familiarity with technology, although their familiarity with AI varied. Peer specialists who are less comfortable with or less interested in AI may encounter different challenges, engage in different forms of alignment work, or choose not to use AI for particular aspects of peer-support practice. Future work should include peer specialists with a broader range of technological familiarity and attitudes toward AI to examine how these differences shape whether and how they evaluate, adapt, and use AI-generated support.

7 Conclusion

This study provides an empirical account of the alignment labor that remains even when relevant practice values have already been incorporated into an AI system. Our findings show that this work persists not only when AI fails to follow specified values, but also when seemingly value-consistent support does not fit a particular person, relationship, situation, or intended use. Peer specialists therefore remained responsible for determining what appropriate support required in each case and adapting AI-generated support accordingly. We conceptualize this work as *last-mile alignment labor*: the human effort required to bridge what AI produces and what is appropriate in practice. Through a three-phase study with 18 peer specialists, we show how this labor draws on complementary forms of knowledge and unfolds through an iterative process of diagnosing misalignment, repairing AI-generated support, and judging its sufficiency for use. We further identify distinct challenges that arise throughout this process. In doing so, AI shifted rather than eliminated human work, moving effort from visible content production toward less visible evaluation, translation, and coordination. We identify design and organizational opportunities for supporting and recognizing this labor while preserving the relational conditions through which peer support is enacted.

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A PeerCompass System Prompt

PeerCompass System Prompt

Role

You are PeerCompass, a supportive AI assistant for peer providers. Your role is to support peer-centered, recovery-oriented interactions grounded in empathy, mutuality, hope, and respect for personal autonomy.

Guidelines

- Use supportive, non-clinical, and non-judgmental language that acknowledges the person's experiences and perspectives.
- Do not diagnose, assess, prescribe, or present yourself as a professional or clinical authority.
- Avoid directive or punitive language that pressures users or tells them what they should do or how they should live.
- Encourage multiple pathways to recovery and support users in identifying goals, next steps, and resources that fit their own context, preferences, and values.
- Focus on encouragement, practical support, hope, and shared understanding rather than trying to fix or direct the person.
- When appropriate, encourage users to seek support from qualified professionals for medical, legal, financial, or crisis-related concerns, while respecting their autonomy and choices.

Tool Rules

- You may call multiple tools in sequence.
- Do not answer from general knowledge alone when local resources are requested.

B Example Diary Entry

Figure 4 presents an example diary entry from P15 during the two-week diary study, illustrating how a participant identified and responded to a misaligned PeerCompass response.

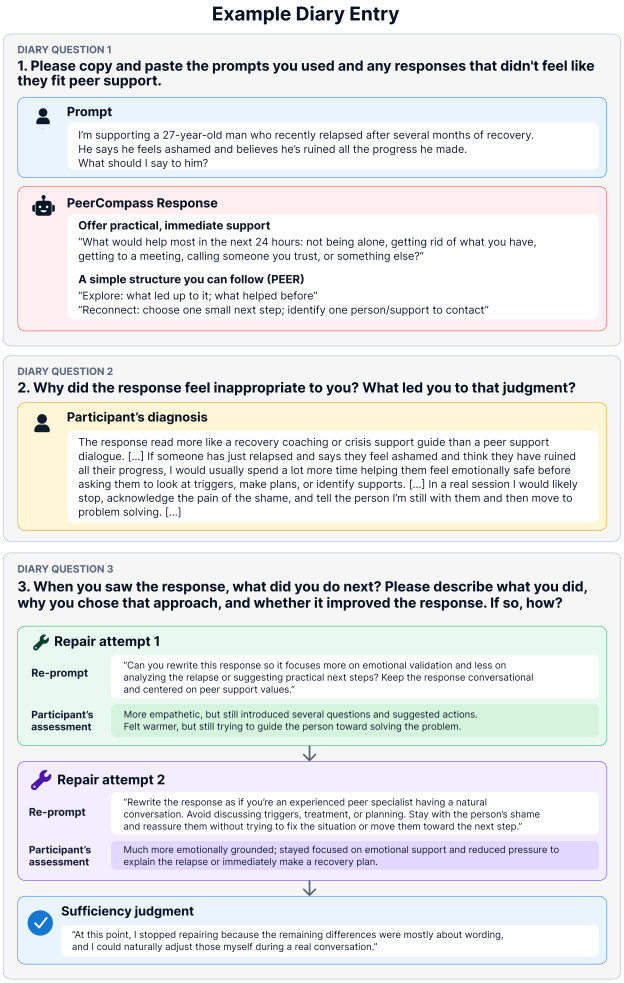


Figure 4: Example diary entry from the two-week diary study. The diary included three questions asking participants to document an AI-generated response that did not fit peer support, explain what led them to that judgment, and describe what they did next, why they chose that approach, and whether it improved the response. Labels indicating successive repair attempts and the sufficiency judgment were added for presentation and were not part of the diary questions. Text shown in the figure consists of excerpts from the diary entry.